In [53]:
```
print("Hello, Titanic EDA!")
```

```
Hello, Titanic EDA!
```

In [54]:
```
import pandas as pd

df = pd.read_csv("train.csv")

df.head()
```

Out[54]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	I
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0

In [55]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    str    
 4   Sex          891 non-null    str    
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    str    
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str    
 11  Embarked     889 non-null    str    
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

In [56]:
```
df.shape
```

Out[56]:
```
(891, 12)
```

In [57]:
```
df.columns
```

Out[57]:
```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='str')
```

In [58]:
```
df.describe()
```

Out[58]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329

In [59]:
```
df.isnull().sum()
```

Out[59]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [60]:
```
df.shape
```

Out[60]:
```
(891, 12)
```

In [61]:
```
df.columns
```

Out[61]:
```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='str')
```

In [62]:
```
df.describe()
```

Out[62]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.00(
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.20
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.69
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.00(
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.91(
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.45
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.00(
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.32

In [63]:
```
df.isnull().sum()
```

Out[63]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [64]:
```
missing_percentage = (df.isnull().sum() / len(df)) * 100
missing_percentage
```

Out[64]:
```
PassengerId     0.000000
Survived        0.000000
Pclass          0.000000
Name            0.000000
Sex             0.000000
Age            19.865320
SibSp           0.000000
Parch           0.000000
Ticket          0.000000
Fare            0.000000
Cabin          77.104377
Embarked        0.224467
dtype: float64
```

Missing Value Observation

The dataset contains missing values in three columns: Age, Cabin, and Embarked. The Cabin column has the highest number of missing values, while Embarked has only two missing values. Age also contains a significant number of missing observations and may require appropriate treatment during data cleaning.

In [65]:
```
df["Survived"].value_counts()
```

Out[65]:
```
Survived
0    549
1    342
Name: count, dtype: int64
```

In [66]:
```
df["Survived"].value_counts(normalize=True) * 100
```

Out[66]:
```
Survived
0    61.616162
1    38.383838
Name: proportion, dtype: float64
```

In [67]:
```
df["Pclass"].value_counts()
```

Out[67]:
```
Pclass
3    491
1    216
2    184
Name: count, dtype: int64
```

In [68]:
```
df["Sex"].value_counts()
```

Out[68]:
```
Sex
male      577
female    314
Name: count, dtype: int64
```

In [69]:
```
df["Embarked"].value_counts()
```

Out[69]:
```
Embarked
S    644
C    168
Q     77
Name: count, dtype: int64
```

Categorical Analysis Observation

- **Survival:** 38.38% of passengers survived, while 61.62% did not survive.
- **Passenger Class:** Most passengers belonged to **3rd class (491)**, followed by 1st class (216) and 2nd class (184).
- **Gender:** The dataset contains **577 male** and **314 female** passengers, so male passengers were more numerous.
- **Embarked:** Most passengers embarked from **Southampton (S) with 644 passengers**, followed by Cherbourg (C) with 168 and Queenstown (Q) with 77.

In [70]:
```
df.duplicated().sum()
```

Out[70]:
```
np.int64(0)
```

In [71]:
```
df["Age"].isnull().sum()
```

Out[71]:
```
np.int64(177)
```

In [72]:
```
df["Age"] = df["Age"].fillna(df["Age"].median())
```

In [73]:
```
df["Age"].isnull().sum()
```

Out[73]:
```
np.int64(0)
```

In [74]:
```
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
```

In [75]:
```
df["Embarked"].isnull().sum()
```

Out[75]:
```
np.int64(0)
```

In [76]:
```
df.drop("Cabin", axis=1, inplace=True)
```

In [77]:
```
df.isnull().sum()
```

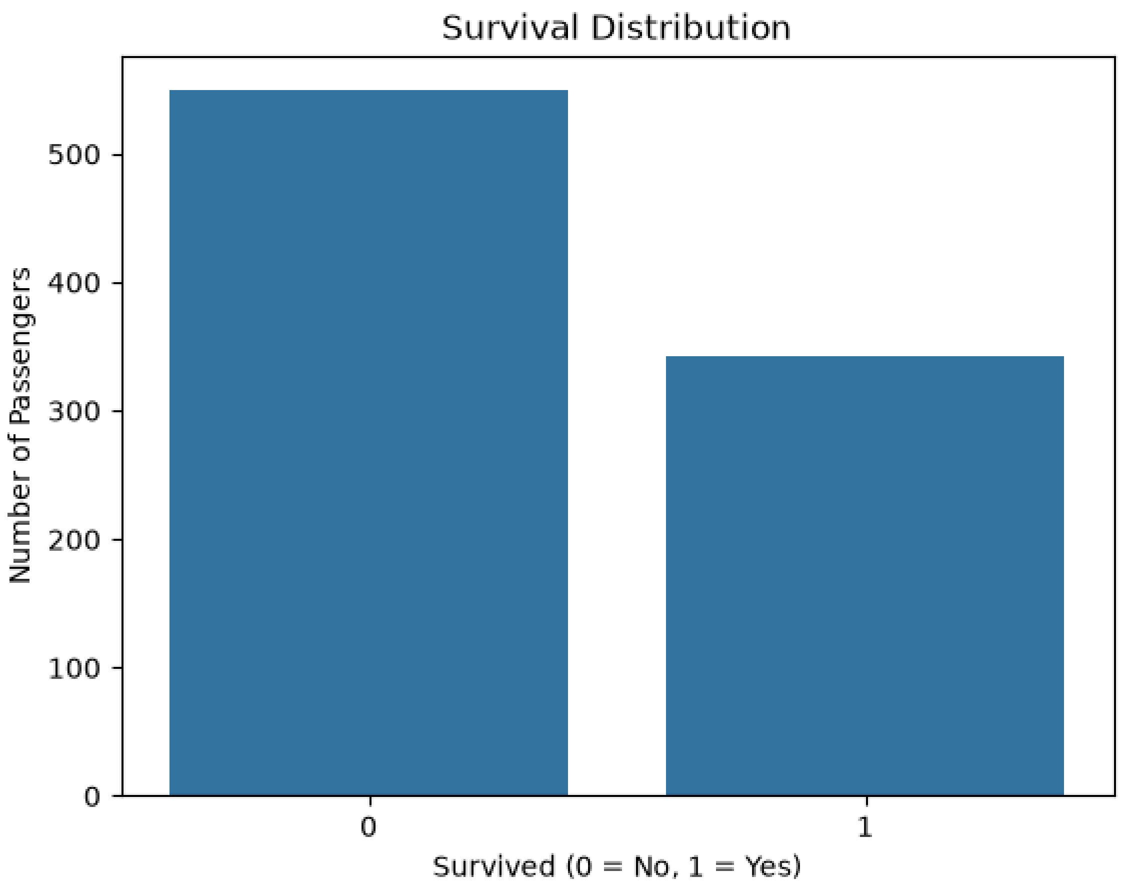
Out[77]:
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

In [78]:
```
%pip install matplotlib seaborn
```

```
Requirement already satisfied: matplotlib in c:\Users\Nitesh\AppData\Local\Progra
ms\Python\Python314\Lib\site-packages (3.11.1)
Requirement already satisfied: seaborn in c:\Users\Nitesh\AppData\Local\Programs
\Python\Python314\Lib\site-packages (0.13.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\Users\Nitesh\AppData\Local
\Programs\Python\Python314\Lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\Users\Nitesh\AppData\Local\Prog
rams\Python\Python314\Lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in c:\Users\Nitesh\AppData\Local
\Programs\Python\Python314\Lib\site-packages (from matplotlib) (4.64.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\Users\Nitesh\AppData\Local
\Programs\Python\Python314\Lib\site-packages (from matplotlib) (1.5.1)
Requirement already satisfied: numpy>=1.25 in c:\Users\Nitesh\AppData\Local\Progr
ams\Python\Python314\Lib\site-packages (from matplotlib) (2.5.2)
Requirement already satisfied: packaging>=20.0 in c:\Users\Nitesh\AppData\Local\P
rograms\Python\Python314\Lib\site-packages (from matplotlib) (26.3)
Requirement already satisfied: pillow>=9 in c:\Users\Nitesh\AppData\Local\Program
s\Python\Python314\Lib\site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in c:\Users\Nitesh\AppData\Local\Prog
rams\Python\Python314\Lib\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in c:\Users\Nitesh\AppData\Lo
cal\Programs\Python\Python314\Lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: pandas>=1.2 in c:\Users\Nitesh\AppData\Local\Progr
ams\Python\Python314\Lib\site-packages (from seaborn) (3.0.5)
Requirement already satisfied: tzdata in c:\Users\Nitesh\AppData\Local\Programs\P
ython\Python314\Lib\site-packages (from pandas>=1.2->seaborn) (2026.3)
Requirement already satisfied: six>=1.5 in c:\Users\Nitesh\AppData\Local\Programs
\Python\Python314\Lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.1
7.0)
Note: you may need to restart the kernel to use updated packages.
```

In [79]:

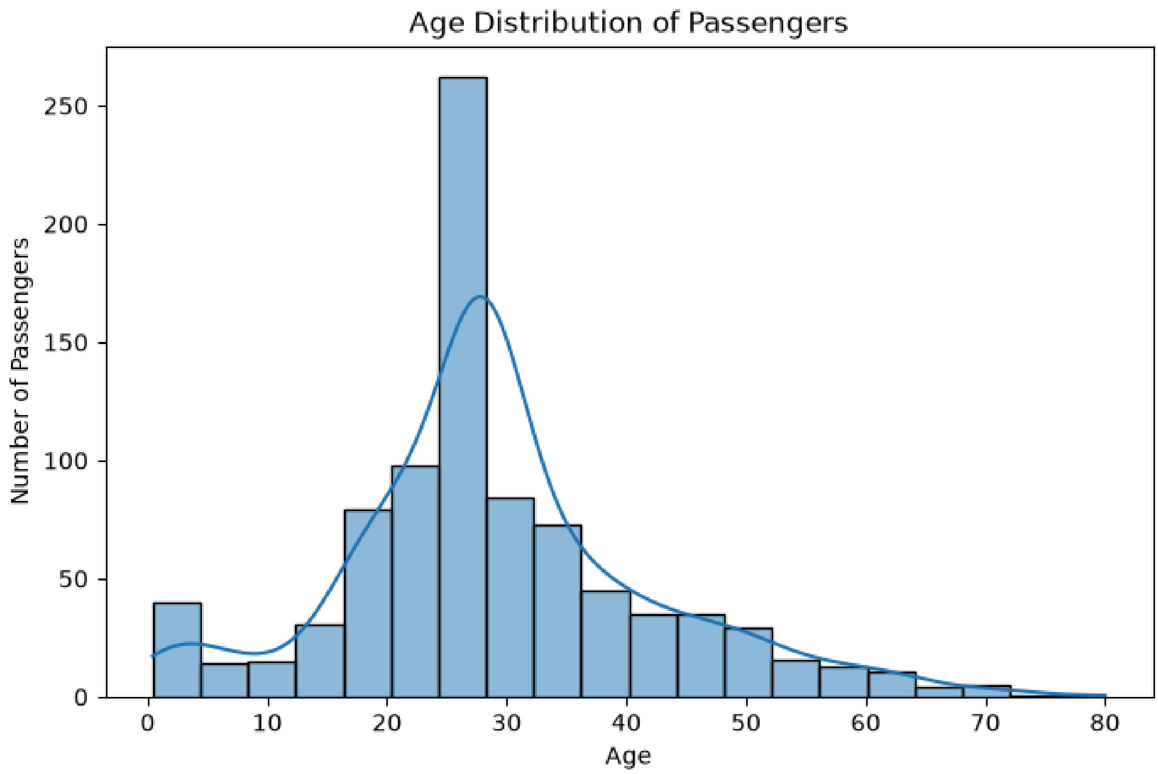
```python
import matplotlib.pyplot as plt
import seaborn as sns

sns.countplot(data=df, x="Survived")
plt.title("Survival Distribution")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of Passengers")
plt.show()
```



Observation — Survival Distribution

The chart shows that the number of passengers who did not survive was higher than the number of passengers who survived. Approximately **61.62% of passengers did not survive**, while **38.38% survived**.

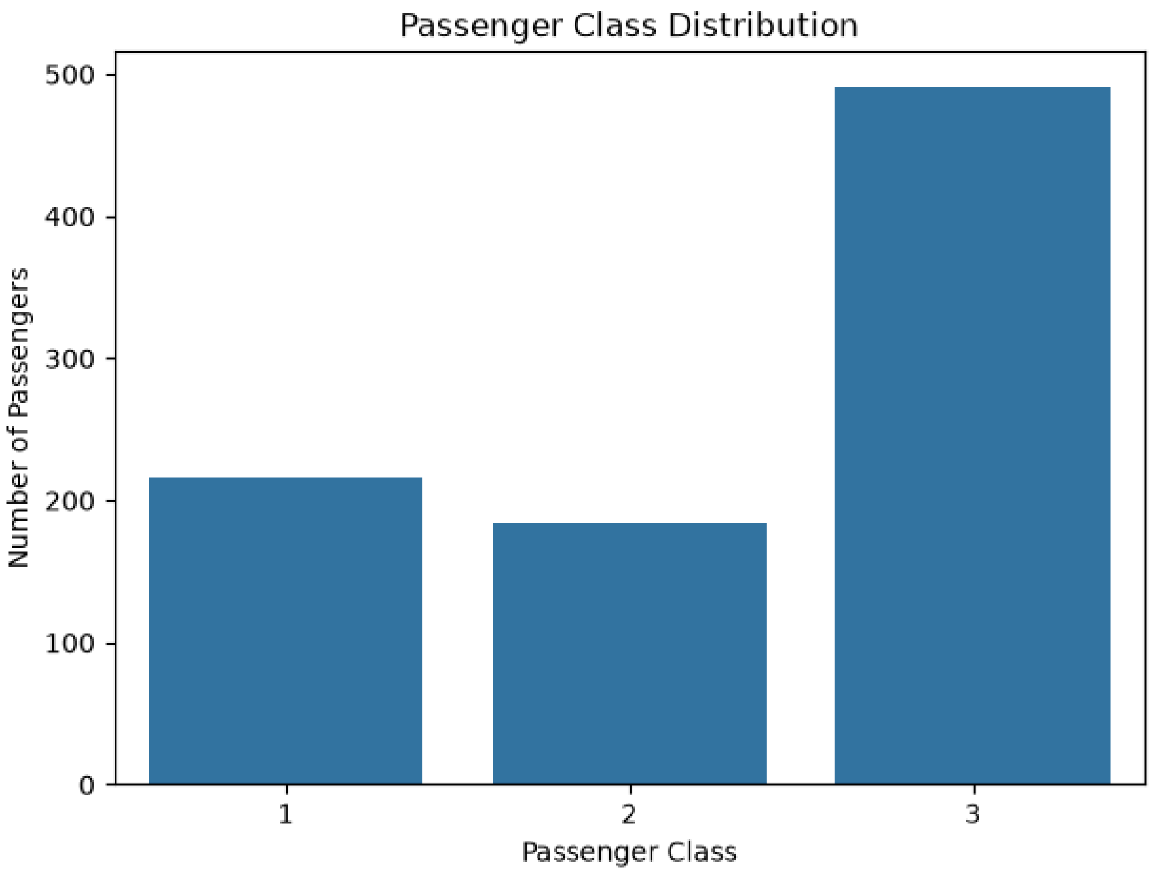
In [80]:
```
plt.figure(figsize=(8, 5))

sns.histplot(data=df, x="Age", bins=20, kde=True)

plt.title("Age Distribution of Passengers")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")
plt.show()
```



Observation — Age Distribution

The age distribution shows that passengers were mostly concentrated in the **young adult and middle-age groups**. The distribution is spread across a wide age range, from infants to older passengers. The highest concentration appears around the **20–40 years** age range.

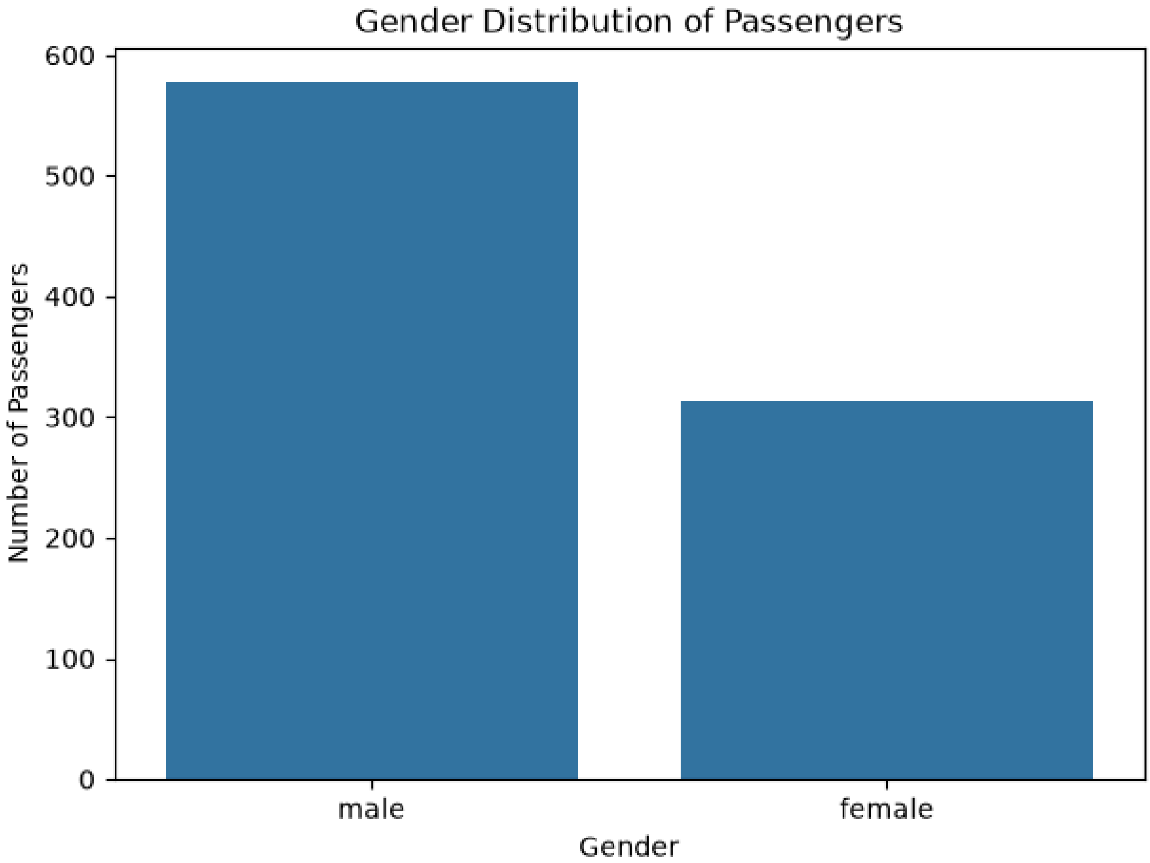
In [81]:
```
plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="Pclass")

plt.title("Passenger Class Distribution")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.show()
```



Observation — Passenger Class Distribution

The chart shows that **3rd class had the highest number of passengers (491)**, followed by **1st class (216)** and **2nd class (184)**. Therefore, most passengers in the Titanic dataset belonged to the 3rd class.

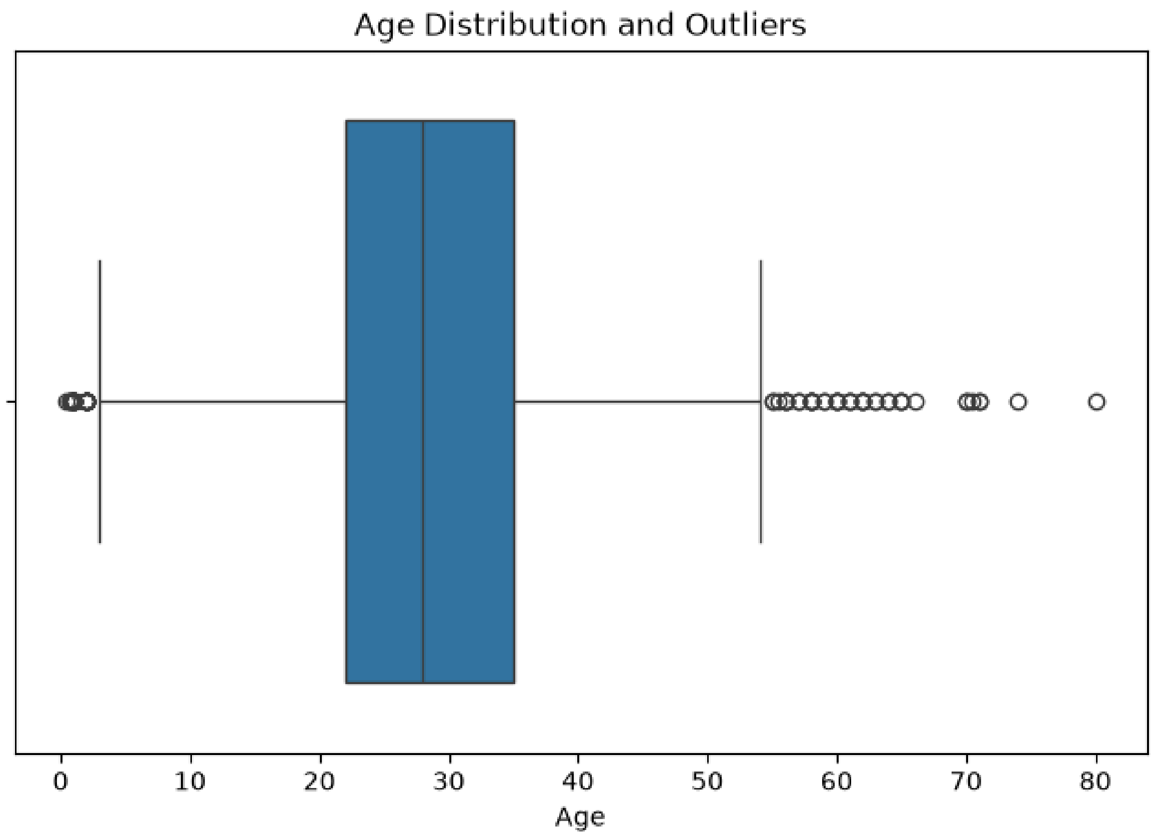
In [82]:
```
plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="Sex")

plt.title("Gender Distribution of Passengers")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.show()
```



Observation — Gender Distribution

The chart shows that the Titanic dataset contains **577 male passengers and 314 female passengers**. Therefore, male passengers were considerably more numerous than female passengers in the dataset.

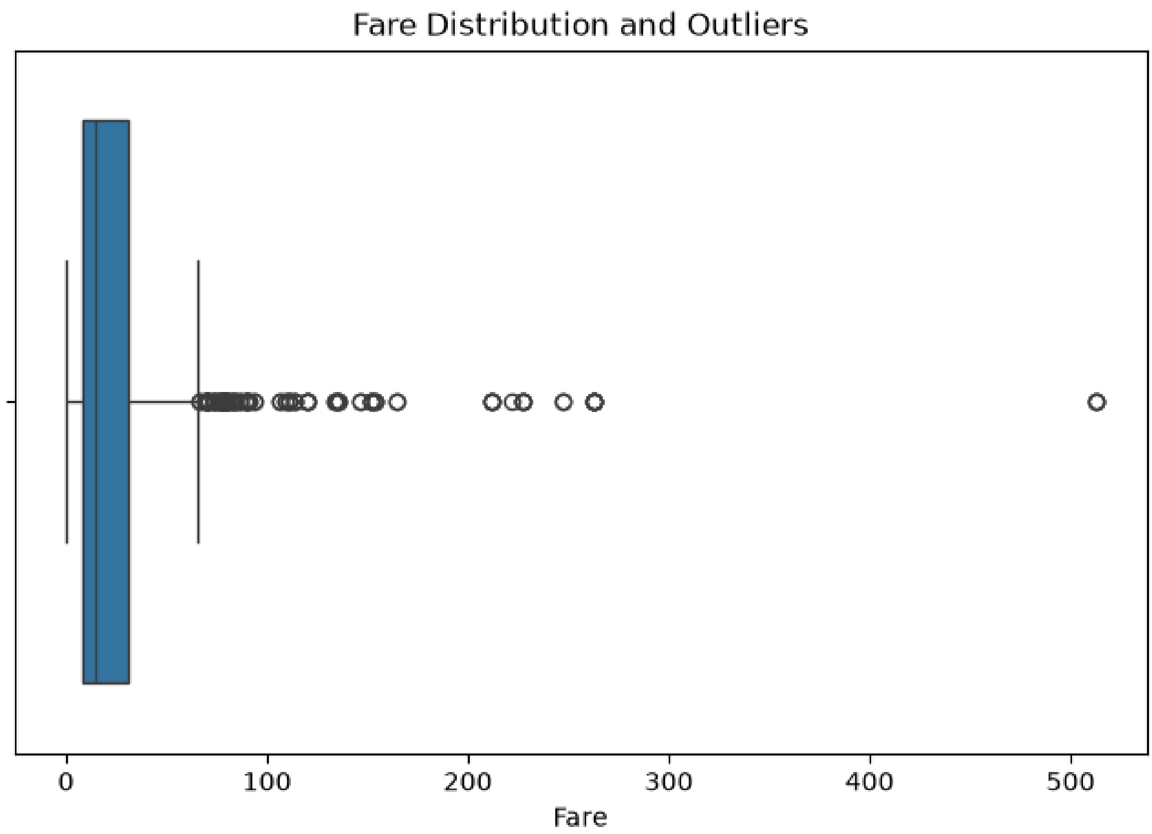
In [83]:
```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Age")

plt.title("Age Distribution and Outliers")
plt.xlabel("Age")
plt.show()
```



Observation — Age Boxplot

The boxplot shows that most passenger ages are concentrated within a relatively normal range. A few passengers are noticeably older than the majority, appearing as potential outliers. However, these values are realistic and should not be removed automatically.

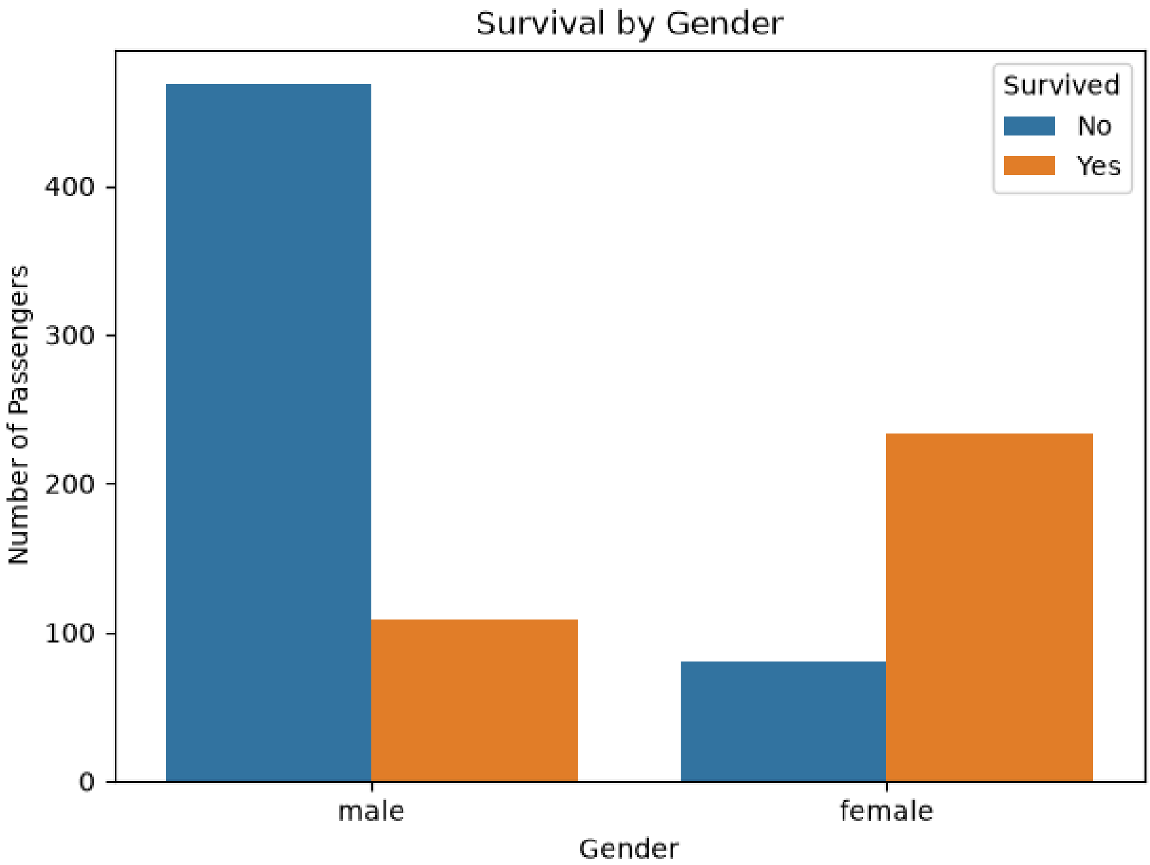
In [84]:
```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Fare")

plt.title("Fare Distribution and Outliers")
plt.xlabel("Fare")
plt.show()
```



Observation — Fare Boxplot

The boxplot shows that most passenger fares were relatively low, while a small number of passengers paid considerably higher fares. These high fare values appear as **potential outliers**. The outliers may be associated with passengers travelling in higher passenger classes or purchasing expensive tickets.

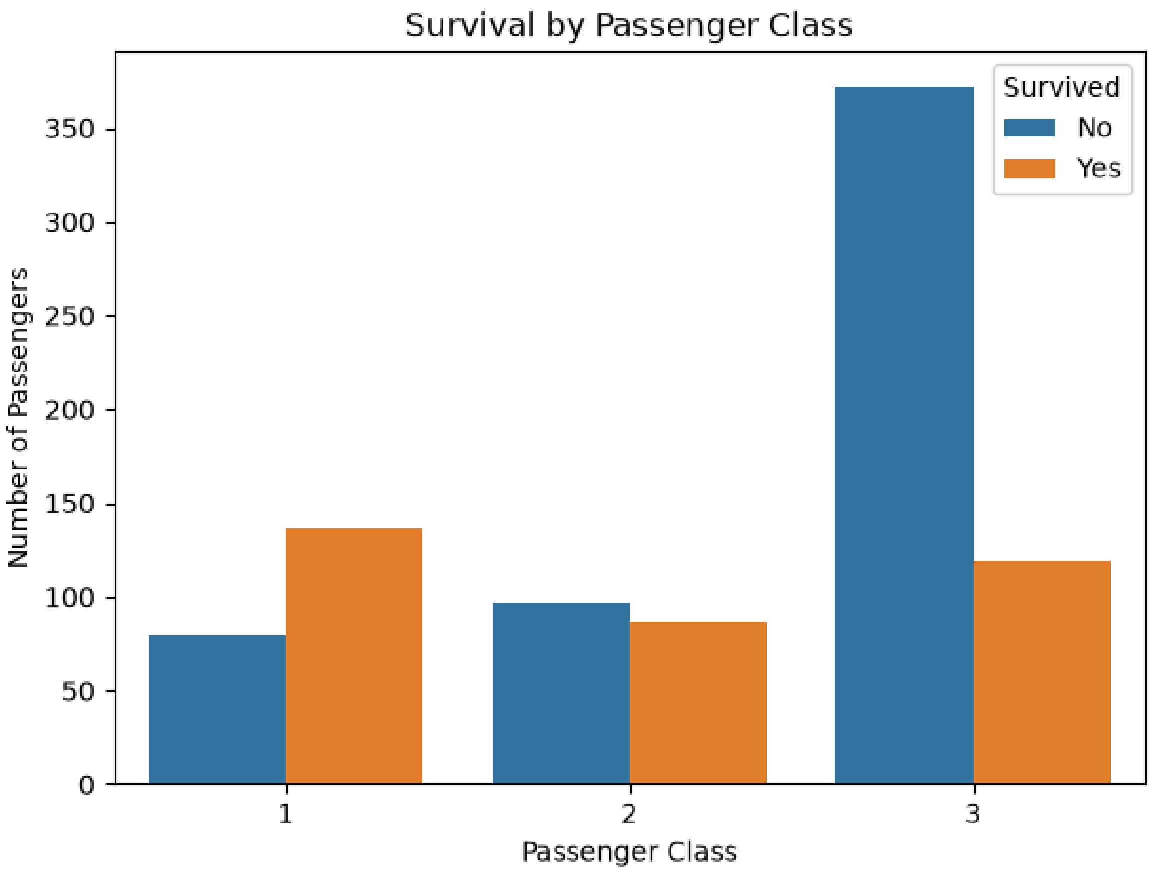
In [85]:
```
plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="Sex", hue="Survived")

plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.legend(title="Survived", labels=["No", "Yes"])
plt.show()
```



Observation — Gender vs Survival

The chart shows a clear difference in survival between male and female passengers. **Female passengers had a much higher survival rate than male passengers**, while a larger proportion of male passengers did not survive. This indicates that gender was an important factor associated with survival on the Titanic.

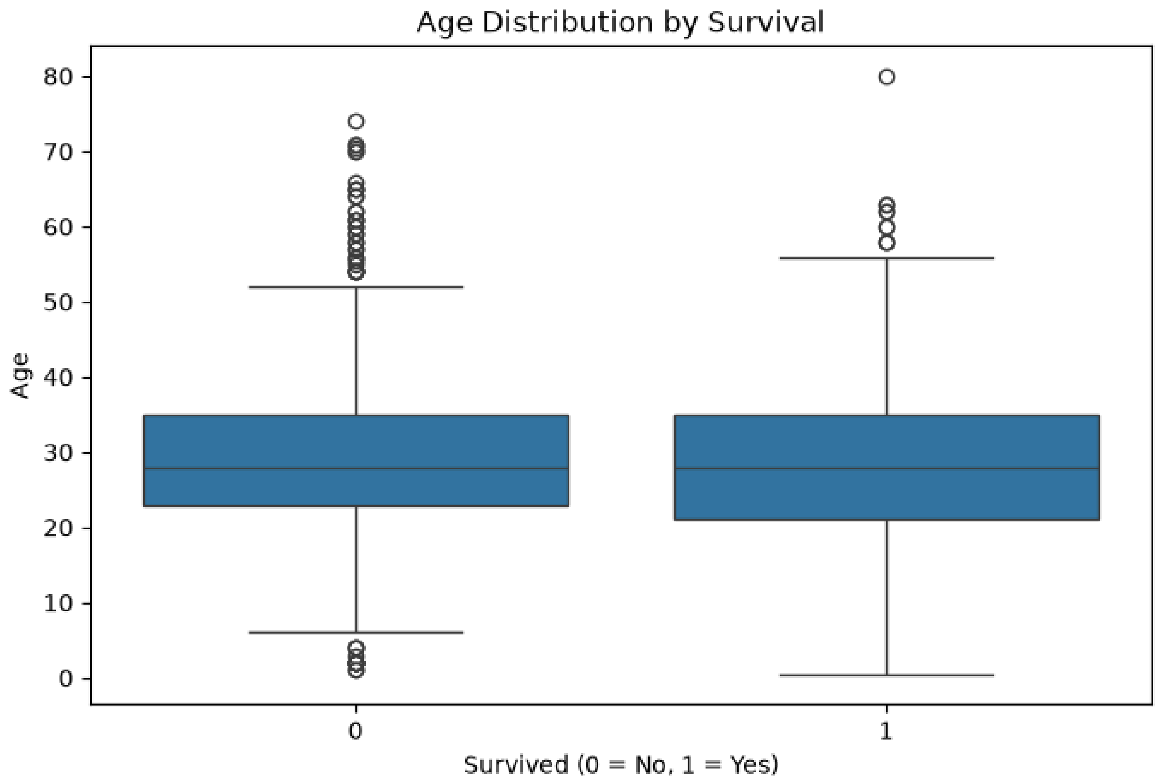
In [86]:
```
plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="Pclass", hue="Survived")

plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.legend(title="Survived", labels=["No", "Yes"])
plt.show()
```



Observation — Passenger Class vs Survival

The chart shows a clear relationship between passenger class and survival. **1st-class passengers had a higher proportion of survivors**, while **3rd-class passengers had the highest number of non-survivors**. Overall, passengers in higher classes were more likely to survive than passengers in lower classes.

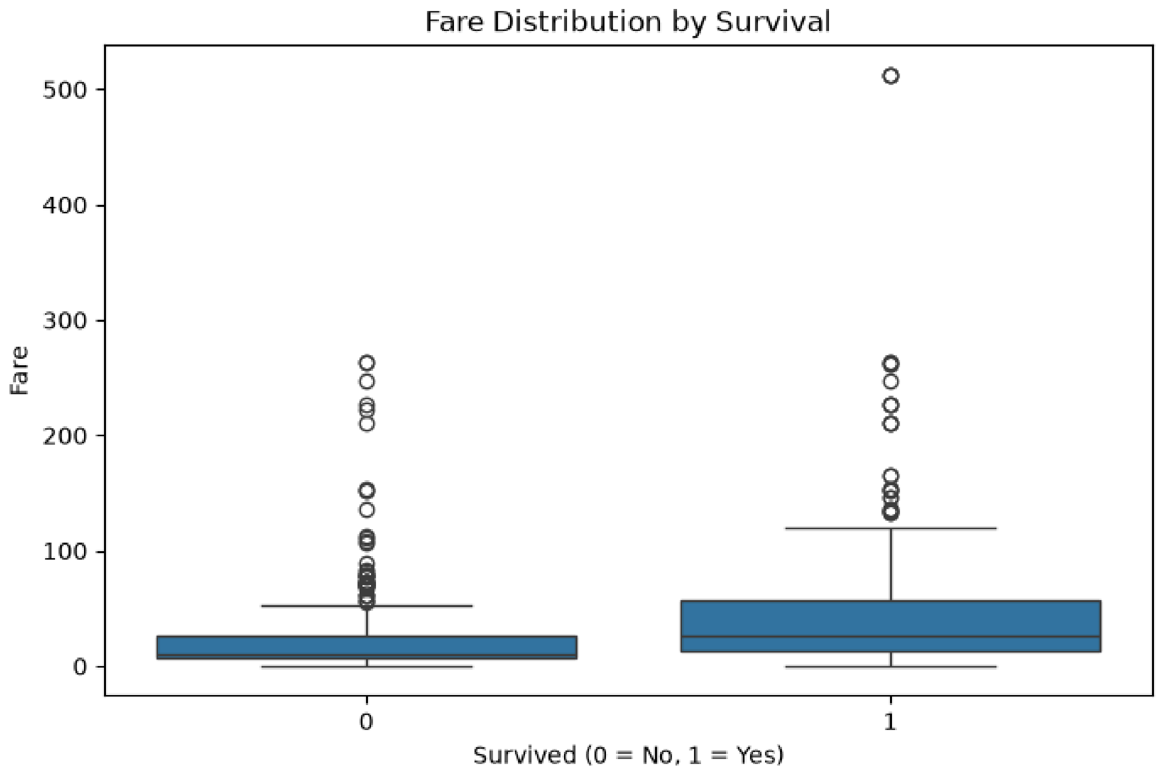
In [87]:
```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Survived", y="Age")

plt.title("Age Distribution by Survival")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Age")
plt.show()
```



Observation — Age vs Survival

The boxplot shows that the age distributions of survivors and non-survivors overlap considerably. However, the survivor group includes a wider range of ages, including younger passengers. This suggests that **age may have influenced survival, but its effect was not as strong as factors such as gender and passenger class**.

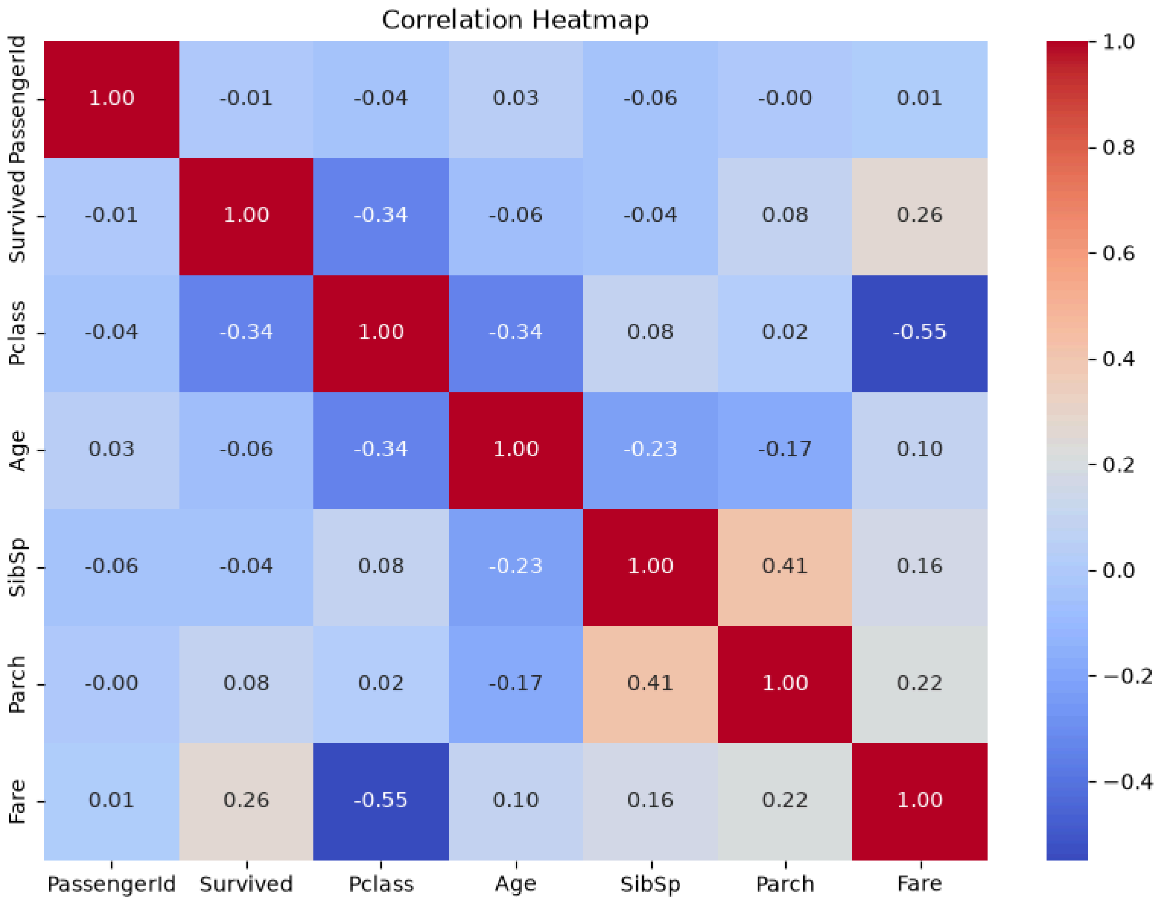
In [88]:
```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Survived", y="Fare")

plt.title("Fare Distribution by Survival")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Fare")
plt.show()
```



Observation — Fare vs Survival

The boxplot shows that passengers who survived generally paid **higher fares** than passengers who did not survive. The survivor group also contains several high-fare values. This suggests that **higher fare levels were associated with a greater likelihood of survival**, which may also be related to passenger class.

In [89]:
```
plt.figure(figsize=(10, 7))

correlation = df.select_dtypes(include="number").corr()

sns.heatmap(correlation, annot=True, cmap="coolwarm", fmt=".2f")

plt.title("Correlation Heatmap")
plt.show()
```



Observation — Correlation Heatmap

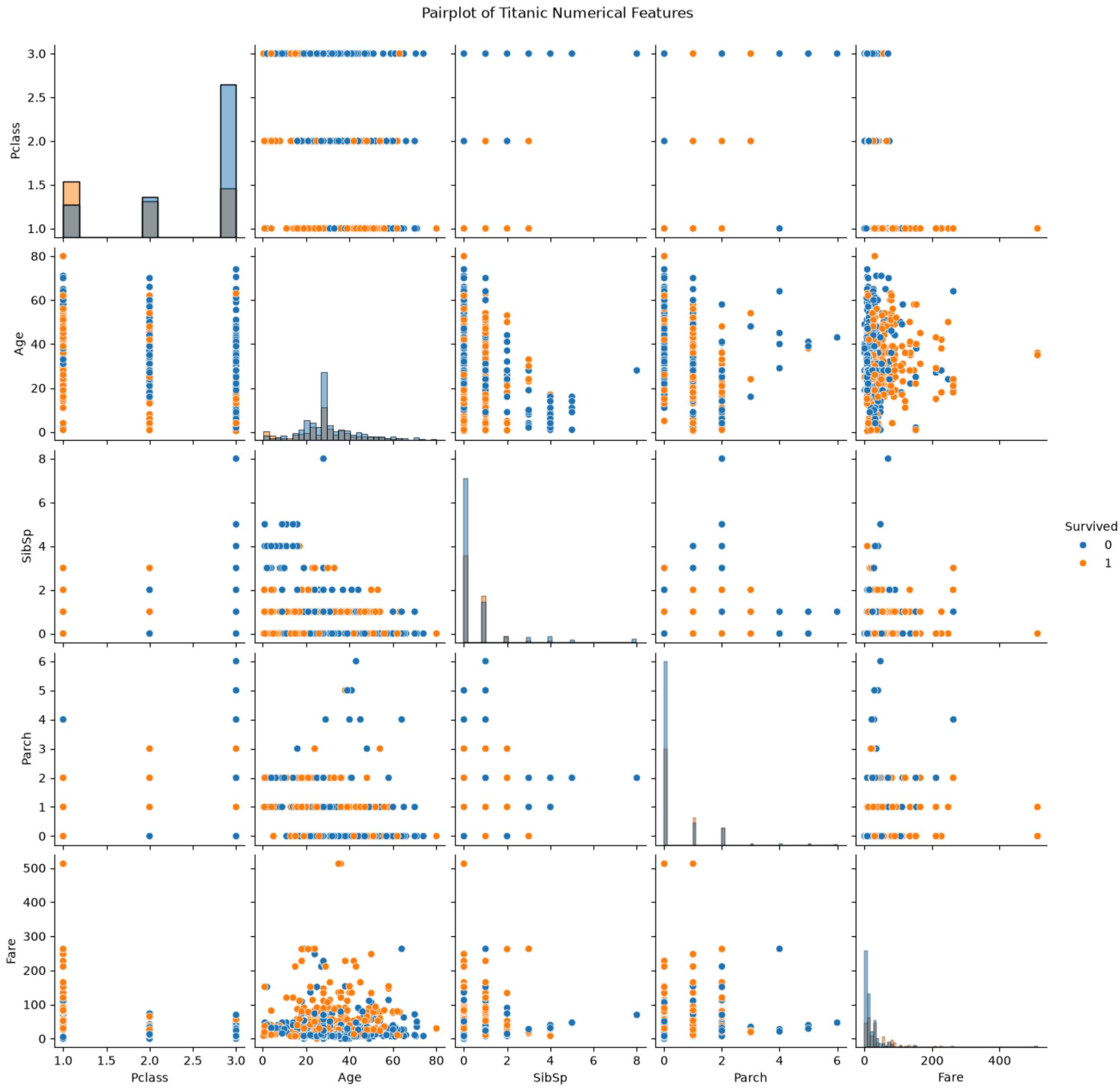
The correlation heatmap shows that **Pclass and Survived have a negative correlation of -0.34**, indicating that passenger class was associated with survival.

Fare and Survived have a positive correlation of 0.26, suggesting that passengers paying higher fares were somewhat more likely to survive.

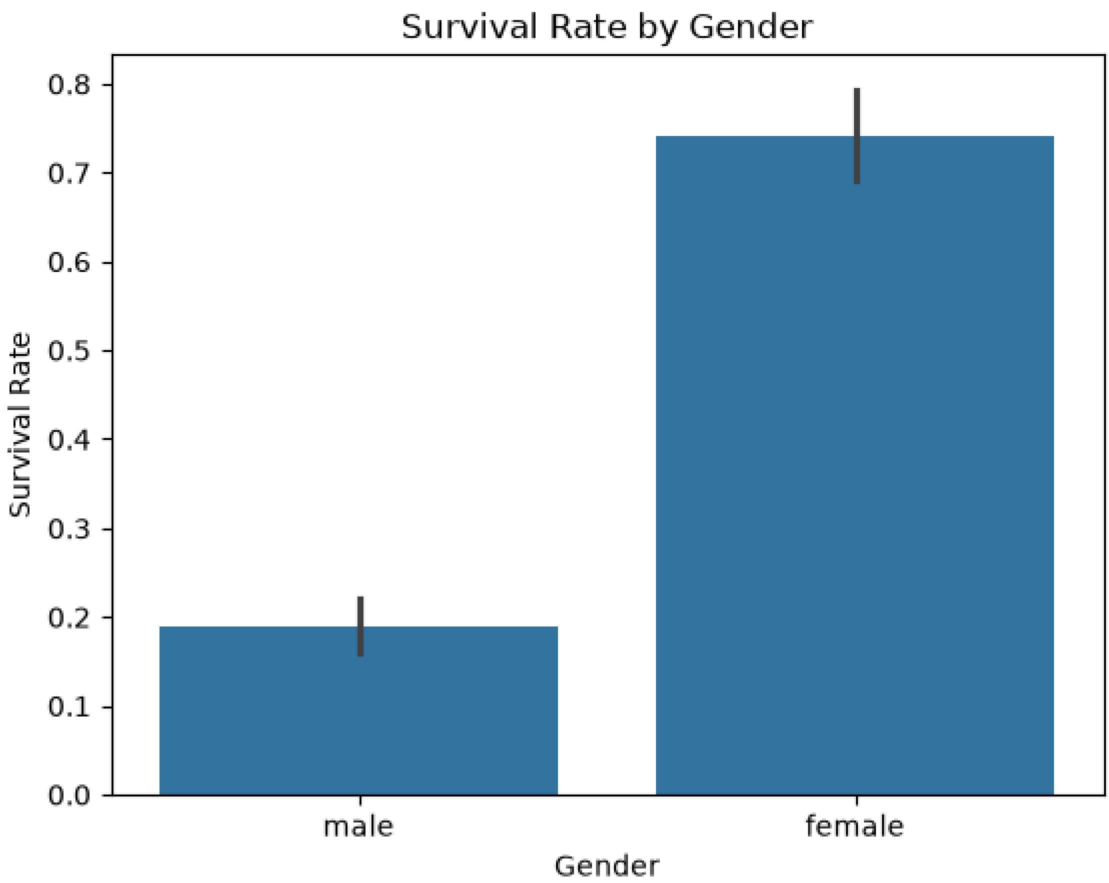
The strongest relationship in the heatmap is between **Pclass and Fare (-0.55)**, showing a clear association between passenger class and fare.

SibSp and Parch have a moderate positive correlation of 0.41, indicating a relationship between the number of siblings/spouses and parents/children travelling with passengers.

In [90]:
```
pairplot_data = df[["Survived", "Pclass", "Age", "SibSp", "Parch", "Fare"]]

sns.pairplot(
    pairplot_data,
    hue="Survived",
    diag_kind="hist"
)

plt.suptitle("Pairplot of Titanic Numerical Features", y=1.02)
plt.show()
```

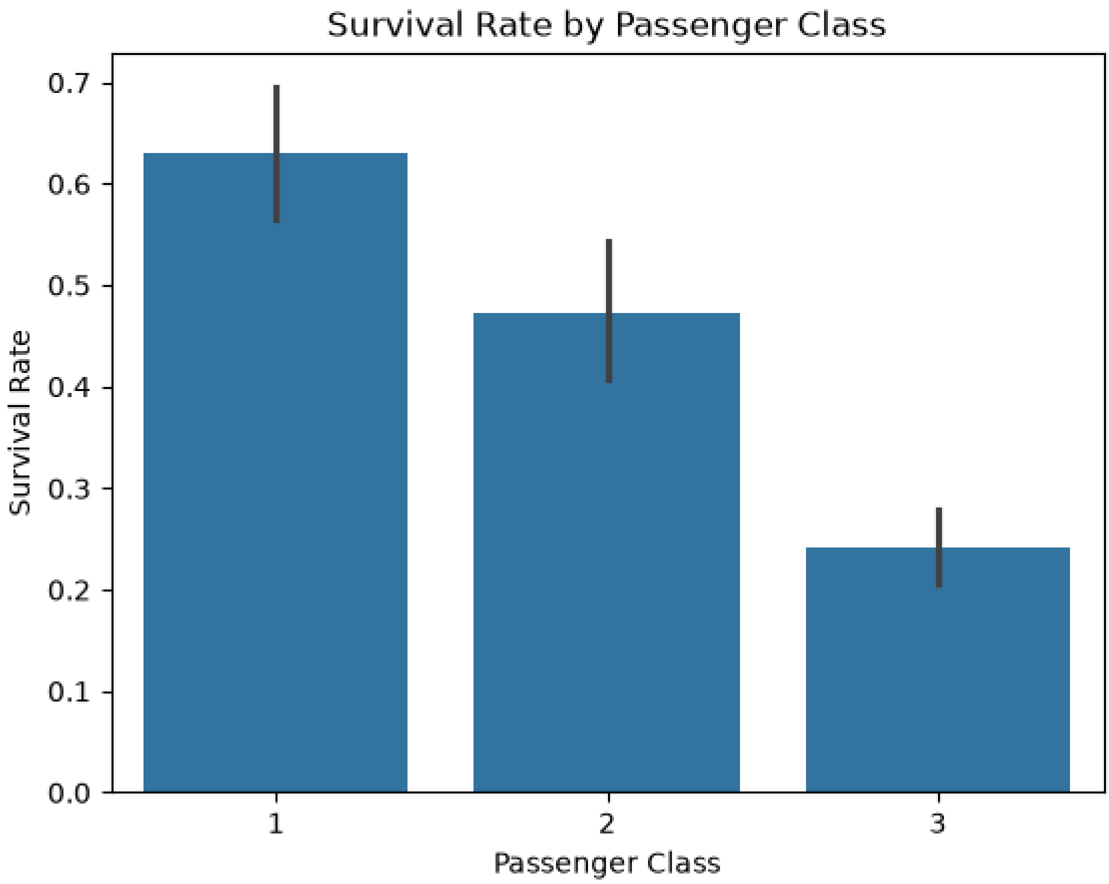


In [91]:
```
sns.barplot(data=df, x='Sex', y='Survived')
plt.title('Survival Rate by Gender')
plt.xlabel('Gender')
plt.ylabel('Survival Rate')
plt.show()
```

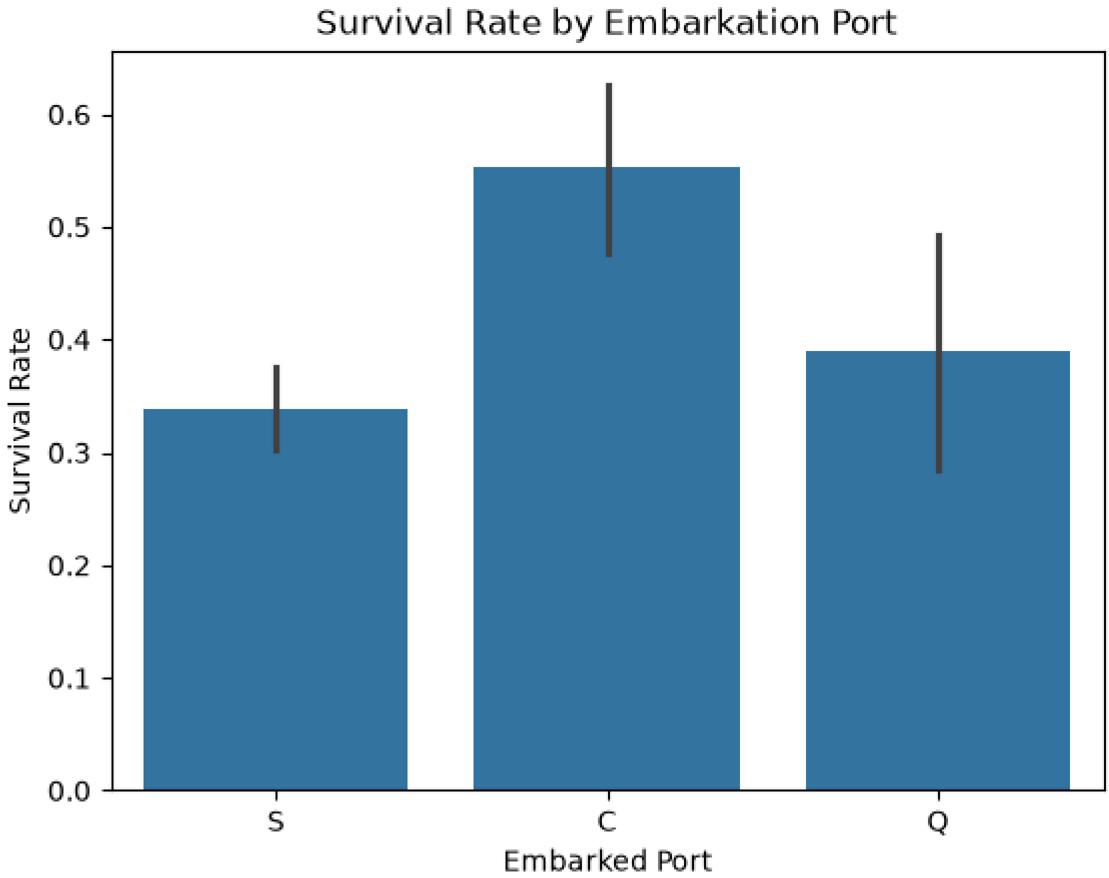


In [92]:

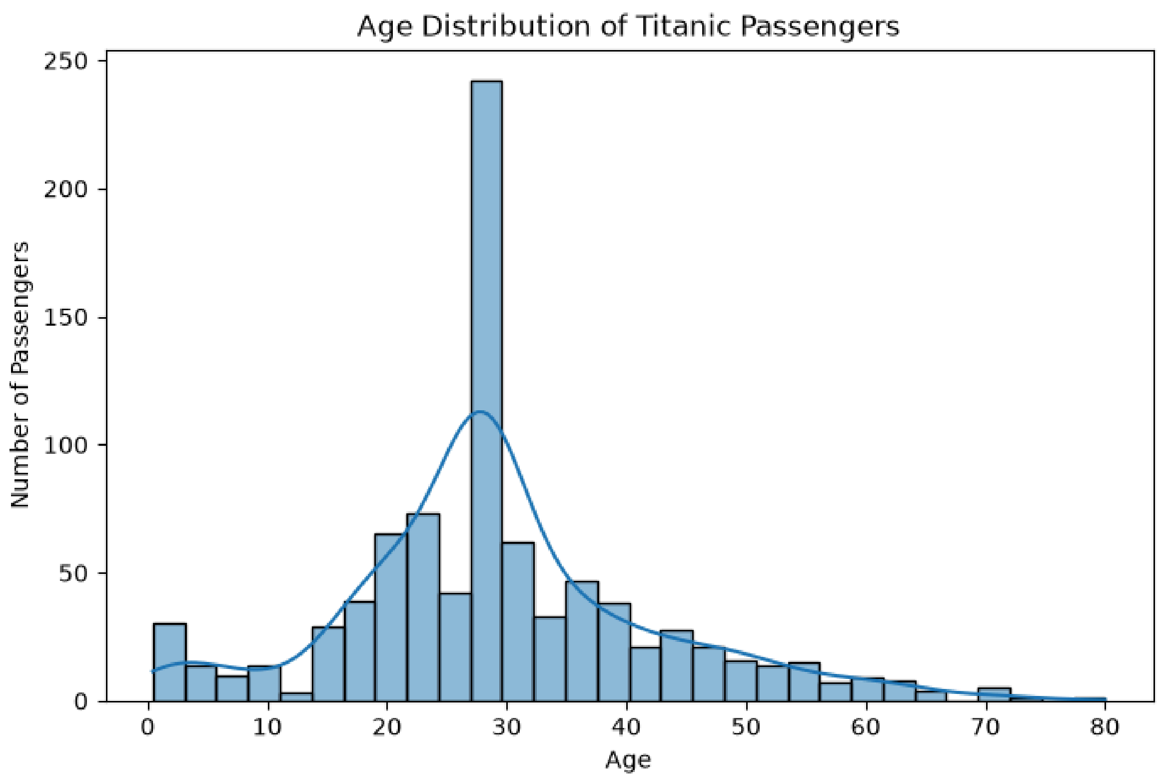
```
sns.barplot(data=df, x='Pclass', y='Survived')
plt.title('Survival Rate by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Survival Rate')
plt.show()
```



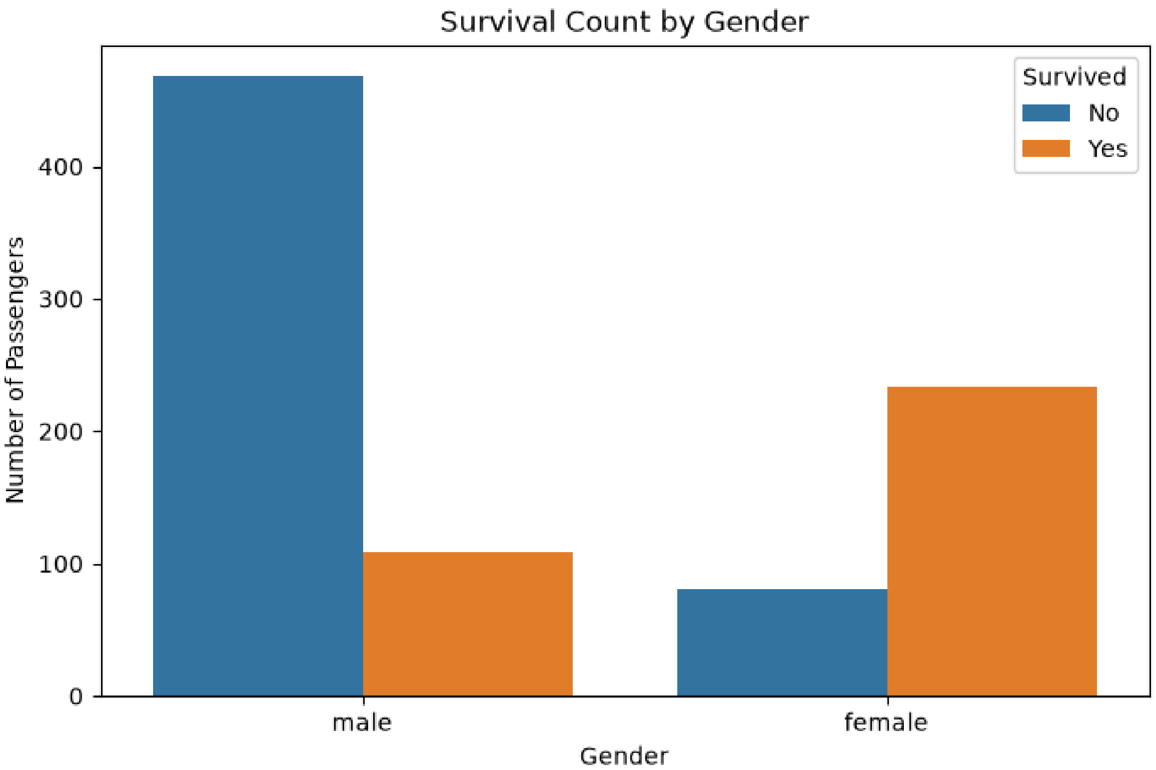
In [93]:
```
sns.barplot(data=df, x='Embarked', y='Survived')
plt.title('Survival Rate by Embarkation Port')
plt.xlabel('Embarked Port')
plt.ylabel('Survival Rate')
plt.show()
```



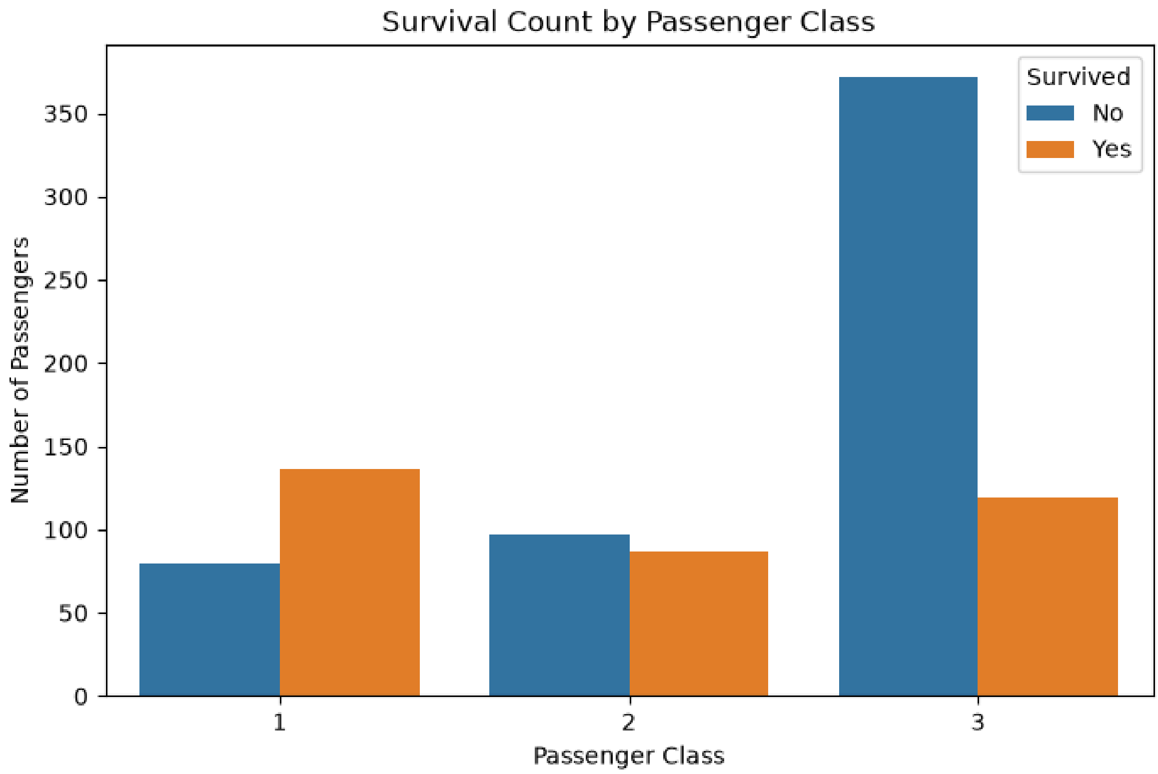
In [94]:
```
plt.figure(figsize=(8, 5))

sns.histplot(data=df, x='Age', bins=30, kde=True)

plt.title('Age Distribution of Titanic Passengers')
plt.xlabel('Age')
plt.ylabel('Number of Passengers')
plt.show()
```



In [95]:
```
plt.figure(figsize=(8, 5))

sns.countplot(data=df, x='Sex', hue='Survived')

plt.title('Survival Count by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()
```

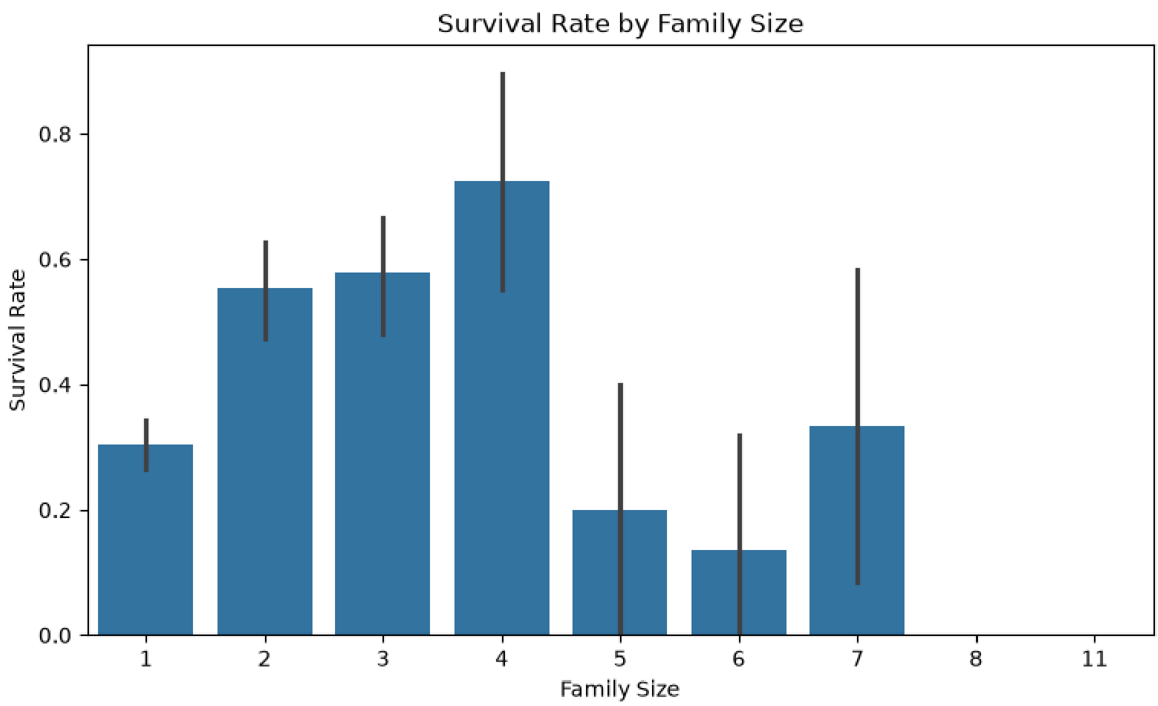


In [96]:
```
plt.figure(figsize=(8, 5))

sns.countplot(data=df, x='Pclass', hue='Survived')

plt.title('Survival Count by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()
```



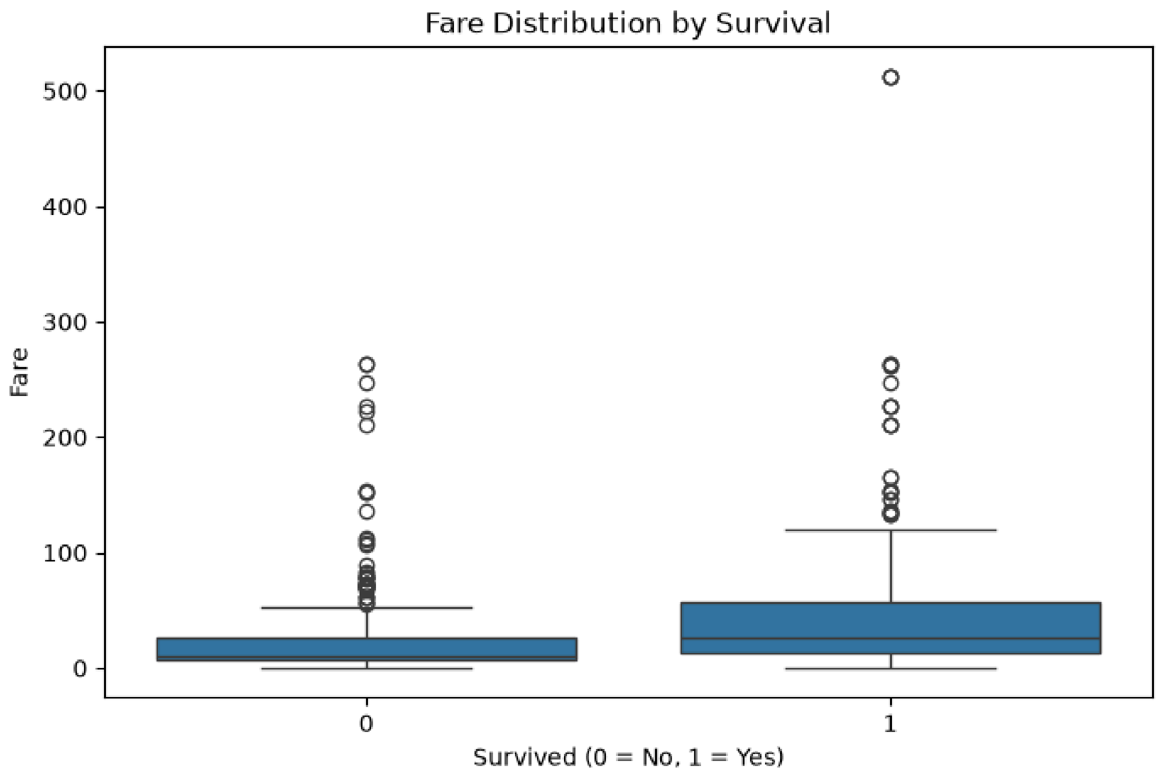
In [97]:
```
df['FamilySize'] = df['SibSp'] + df['Parch'] + 1
```

In [98]:
```
plt.figure(figsize=(9, 5))

sns.barplot(data=df, x='FamilySize', y='Survived')

plt.title('Survival Rate by Family Size')
plt.xlabel('Family Size')
plt.ylabel('Survival Rate')
plt.show()
```



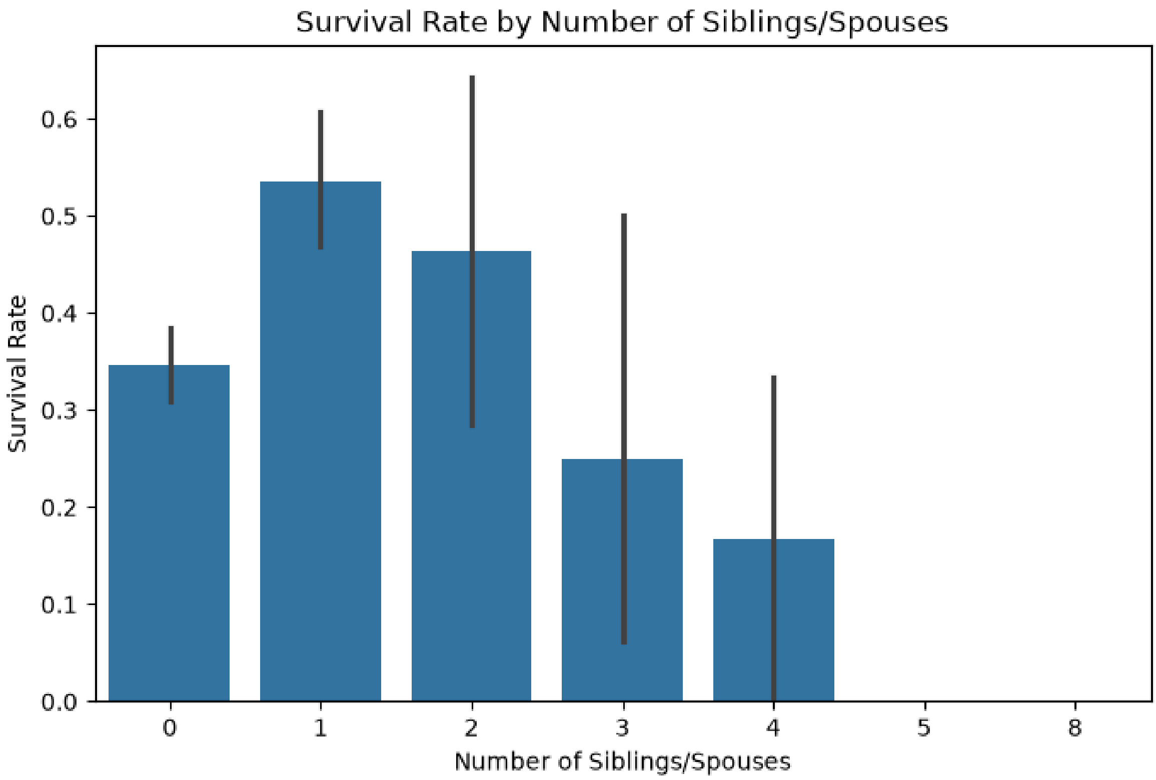
In [99]:
```
plt.figure(figsize=(8, 5))
```

```
sns.boxplot(data=df, x='Survived', y='Fare')

plt.title('Fare Distribution by Survival')
plt.xlabel('Survived (0 = No, 1 = Yes)')
plt.ylabel('Fare')
plt.show()
```



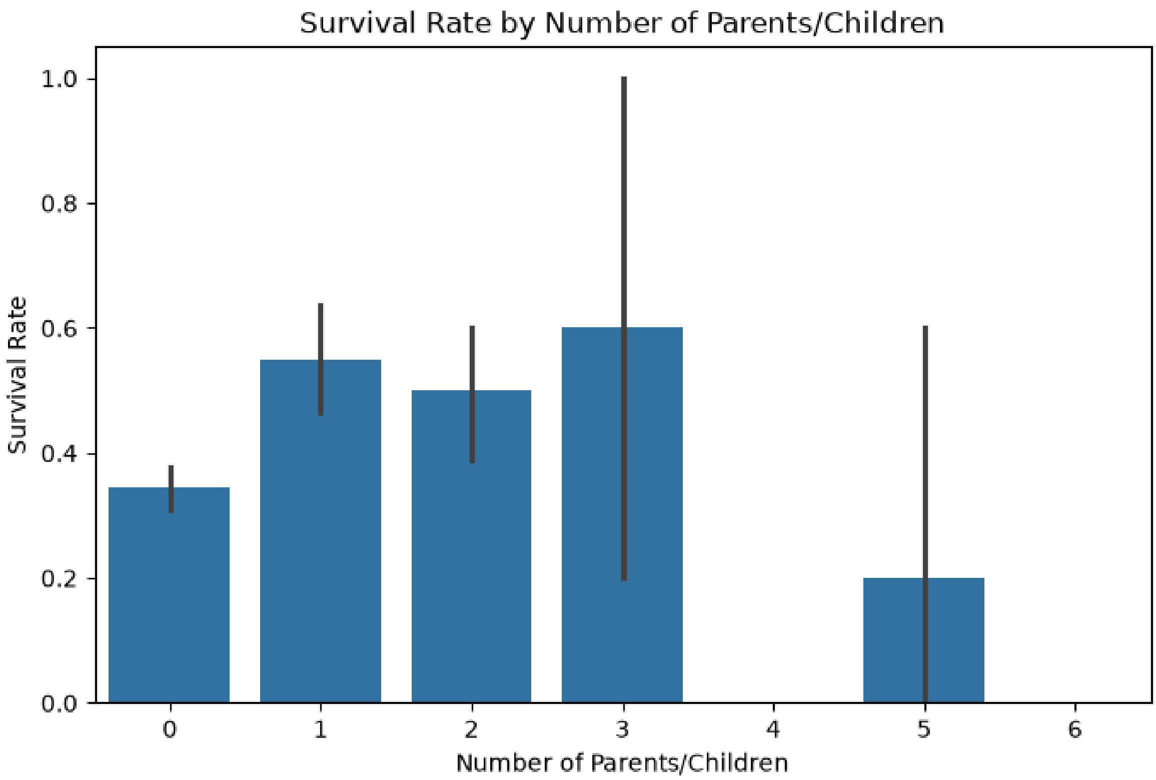
In [100…

```
plt.figure(figsize=(8, 5))

sns.barplot(data=df, x='SibSp', y='Survived')

plt.title('Survival Rate by Number of Siblings/Spouses')
plt.xlabel('Number of Siblings/Spouses')
plt.ylabel('Survival Rate')
plt.show()
```



In [101…

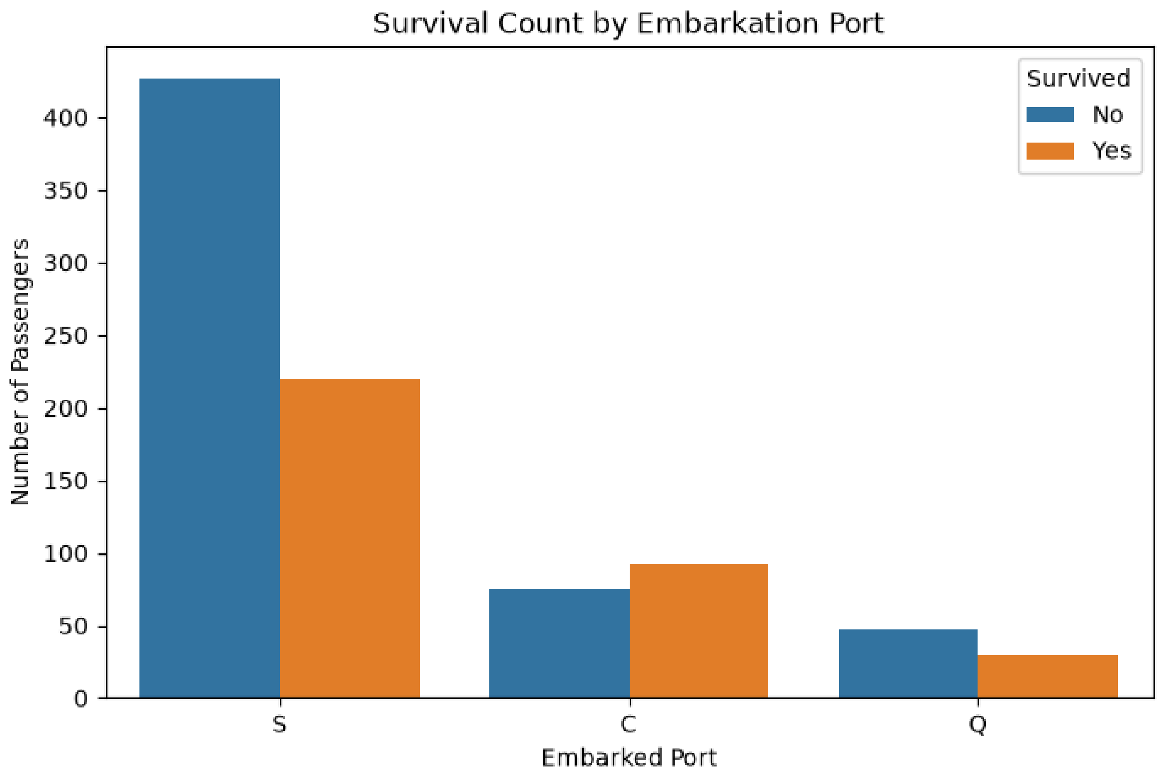
```
plt.figure(figsize=(8, 5))

sns.barplot(data=df, x='Parch', y='Survived')

plt.title('Survival Rate by Number of Parents/Children')
plt.xlabel('Number of Parents/Children')
plt.ylabel('Survival Rate')
plt.show()
```



In [102…

```
plt.figure(figsize=(8, 5))
```

```
sns.countplot(data=df, x='Embarked', hue='Survived')

plt.title('Survival Count by Embarkation Port')
plt.xlabel('Embarked Port')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()
```



In [103…

```
print("Overall Survival Rate:", round(df['Survived'].mean() * 100, 2), "%")

print("\nSurvival Rate by Gender:")
print((df.groupby('Sex')['Survived'].mean() * 100).round(2))

print("\nSurvival Rate by Passenger Class:")
print((df.groupby('Pclass')['Survived'].mean() * 100).round(2))

print("\nSurvival Rate by Embarked Port:")
print((df.groupby('Embarked')['Survived'].mean() * 100).round(2))
```

```
Overall Survival Rate: 38.38 %

Survival Rate by Gender:
Sex
female    74.20
male      18.89
Name: Survived, dtype: float64

Survival Rate by Passenger Class:
Pclass
1    62.96
2    47.28
3    24.24
Name: Survived, dtype: float64

Survival Rate by Embarked Port:
Embarked
C    55.36
Q    38.96
S    33.90
Name: Survived, dtype: float64
```

Key Insights / Findings

1. **Gender:** Female passengers had a significantly higher survival rate than male passengers.

2. **Passenger Class:** 1st-class passengers had the highest survival rate, while 3rd-class passengers had the lowest.

3. **Age:** Passenger age showed differences between survivors and non-survivors, with children having comparatively better survival chances.

4. **Fare:** Passengers who paid higher fares generally had better survival rates, which is related to passenger class.

5. **Family Size:** Survival probability varied with family size. Passengers travelling with a small family generally showed better survival patterns than those travelling alone or with very large families.

6. **Embarkation Port:** Survival rates differed across embarkation ports, with passengers embarking from different ports showing different survival patterns.

7. **Overall Finding:** Gender and passenger class were among the strongest factors associated with survival on the Titanic.

Conclusion

The Titanic dataset was analyzed using Python, Pandas, Matplotlib, and Seaborn. The analysis explored passenger demographics, survival patterns, passenger class, gender, age, fare, family size, and embarkation port.

The main finding is that survival was strongly associated with factors such as gender and passenger class. Female passengers and 1st-class passengers generally had higher chances of survival. Age, fare, and family-related variables also showed noticeable patterns.

Overall, the EDA helped identify important factors associated with passenger survival and demonstrated how data visualization and statistical exploration can be used to extract meaningful insights from a real-world dataset.